

CS1632: Test-Driven Development

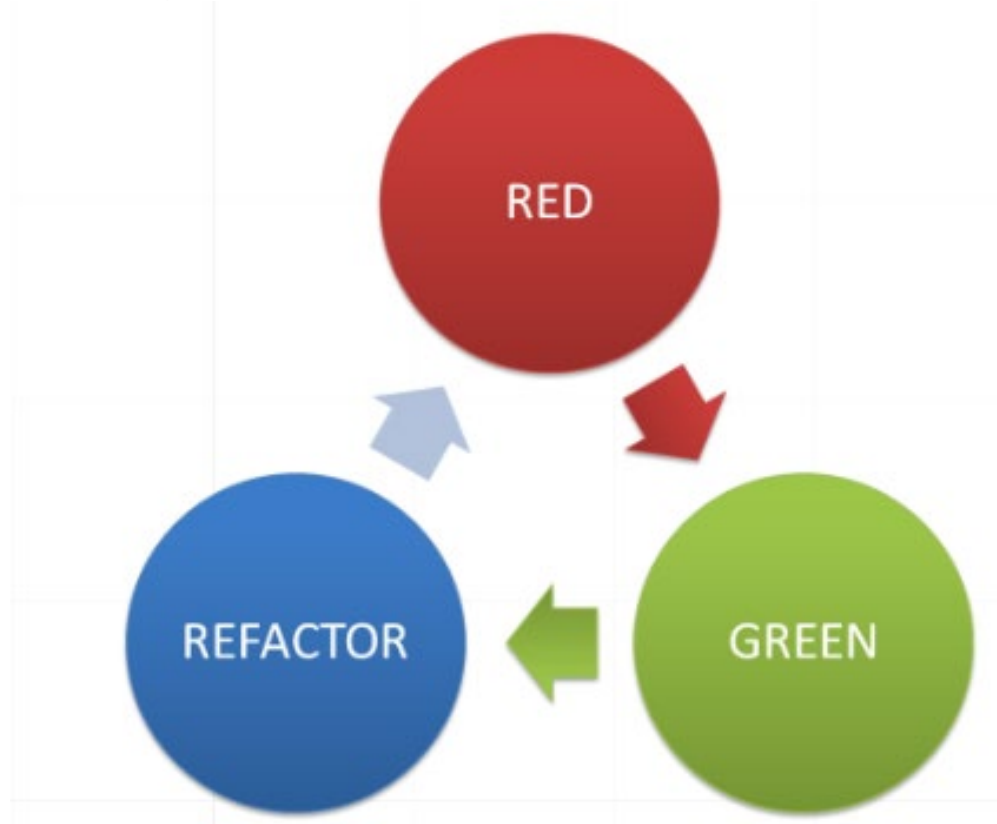
Wonsun Ahn

TDD (Test-Driven Development)

- Remember **Requirements** drives → **Testing** drives → **Development**?
 - Write tests to verify the requirements (traceability matrix)
 - Write code to pass the tests (TDD)
- TDD is a development methodology – meant for developers
 - Developers write the test *before* writing the code

The Red-Green-Refactor Loop

- TDD is performed using the RGR Loop:



The Red-Green-Refactor Loop

- Red – Write a test for a new behavior
 - Test should immediately fail! (Hence the Red)
- Green – Write only enough code to make the test pass
 - Now the test should pass. (Hence the Green)
- Refactor – Review code and make it better

What is Refactoring?

- Even if your code is defect-free, it may not be perfect
 - Poor algorithm choice?
 - Bad variable names?
 - Poor performance?
 - Badly documented?
 - Magic numbers?
 - Not easily comprehensible?
 - General bad design?
- *Refactoring*: improving code without changing its functionality

Fizzbuzzin' With TDD

- Requirements:
 - App shall print numbers from 1 to 100 delimited by spaces.
 - If number is a multiple of 3, app shall print "Fizz" instead.
 - If number is a multiple of 5, app shall print "Buzz" instead.
 - If number is a multiple of 3 and 5, app shall print "FizzBuzz" instead.
 - If number is neither a multiple of 3 and 5, app shall print the actual number.
- Example output: 1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz ...

Red - Start by adding a new test

```
@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}

// Code
public String value(int n) {
    return "";
}
```

Red – Which fails



Green - Write code to make test pass

```
@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}
// Code
public String value(int n) {
    return "1";
}
```

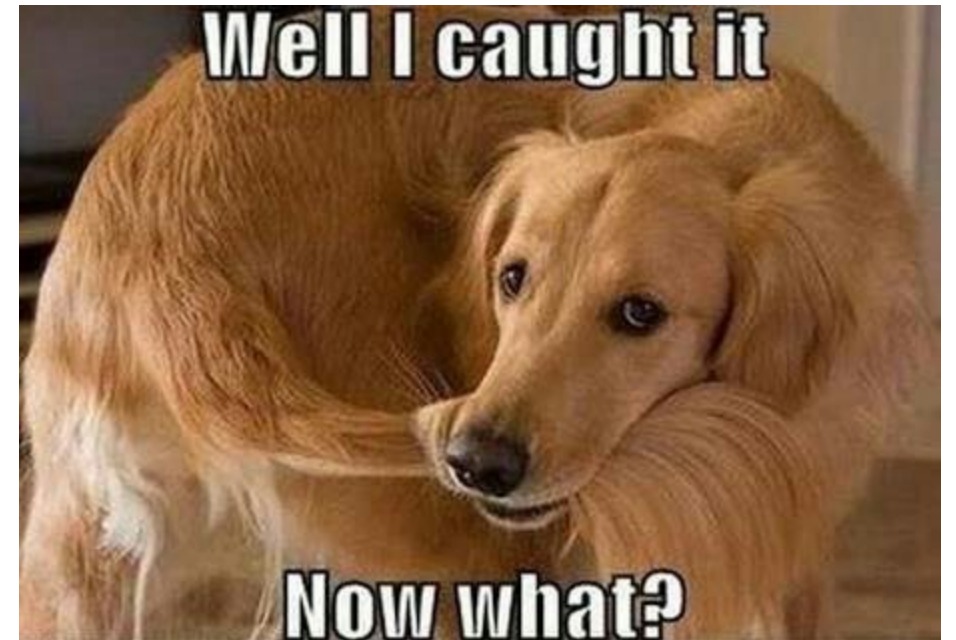
Green – Test passes!

```
@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}
// Code
public String value(int n) {
    return "1";
}
```



Refactor – Nothing to do

```
@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}
// Code
public String value(int n) {
    return "1";
}
```



Red – Well, let's Add Another Test for "2"

```
@Test
public void testNumber2() {
    assertEquals(fb.value(2), "2");
}
```

```
@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}
```

```
// Code
public String value(int n) {
    return "1";
}
```

Red – Which fails (again)



Green - Let's Make Another Change

```
@Test
public void testNumber2() {
    assertEquals(fb.value(2), "2");
}

@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}

// Code
public String value(int n) {
    if (n == 1) {
        return "1";
    } else {
        return "2";
    }
}
```

Green – Tests pass once more!

```
@Test
public void testNumber2() {
    assertEquals(fb.value(2), "2");
}

@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}

// Code
public String value(int n) {
    if (n == 1) {
        return "1";
    } else {
        return "2";
    }
}
```



Refactor – Much Nicer and Tests Still Pass!

```
@Test
public void testNumber2() {
    assertEquals(fb.value(2), "2");
}

@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}

// Code
public String value(int n) {
    return String.valueOf(n);
}
```

Red – Add another test for “3”, which fails

```
@Test
public void testNumber3() {
    assertEquals(fb.value(3), "Fizz");
}

@Test
public void testNumber2() {
    assertEquals(fb.value(2), "2");
}

@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}

// Code, omitted for brevity
```

Green – Add fizzy code to make test pass!

```
private boolean fizzy(int n) {  
    return (n % 3 == 0);  
}
```

```
public String value(int n) {  
    if (fizzy(n)) {  
        return "Fizz";  
    } else {  
        return String.valueOf(n);  
    }  
}
```

Green – Test suite passes again!

```
private boolean fizzy(int n) {  
    return (n % 3 == 0);  
}
```

```
public String value(int n) {  
    if (fizzy(n)) {  
        return "Fizz";  
    } else {  
        return String.valueOf(n);  
    }  
}
```



Refactor – Nothing to do here

```
private boolean fizzy(int n) {  
    return (n % 3 == 0);  
}
```

```
public String value(int n) {  
    if (fizzy(n)) {  
        return "Fizz";  
    } else {  
        return String.valueOf(n);  
    }  
}
```

Red – Let's add a test for buzziness (fails)

```
@Test
public void testNumber5() {
    assertEquals(fb.value(5), "Buzz");
}

@Test
public void testNumber3() {
    assertEquals(fb.value(3), "Fizz");
}

@Test
public void testNumber2() {
    assertEquals(fb.value(2), "2");
}

@Test
public void testNumber() {
    assertEquals(fb.value(1), "1");
}

// Code, omitted for brevity
```

Green – Add buzzy code to make test pass!

```
private boolean buzzy(int n) {  
    return (n % 5 == 0);  
}  
public String value(int n) {  
    if (fizzy(n)) {  
        return "Fizz";  
    } else if (buzzy(n)) {  
        return "Buzz";  
    } else {  
        return String.valueOf(n);  
    }  
}
```

Green – Now all tests pass once more!

```
private boolean buzzy(int n) {  
    return (n % 5 == 0);  
}  
public String value(int n) {  
    if (fizzy(n)) {  
        return "Fizz";  
    } else if (buzzy(n)) {  
        return "Buzz";  
    } else {  
        return String.valueOf(n);  
    }  
}
```



Red - The final equivalence class “FizzBuzz”

```
@Test  
public void testNumber15() {  
    assertEquals(fb.value(15), "FizzBuzz");  
}
```

```
@Test  
public void testNumber5() {  
    assertEquals(fb.value(5), "Buzz");  
}
```

```
@Test  
public void testNumber3() {  
    assertEquals(fb.value(3), "Fizz");  
}
```

```
@Test  
public void testNumber2() {  
    assertEquals(fb.value(2), "2");  
}
```

```
@Test  
public void testNumber() {  
    assertEquals(fb.value(1), "1");  
}
```

Green – Add code to make input 15 pass too!

```
public String value(int n) {  
    if (fizzy(n) && buzzy(n)) {  
        return "FizzBuzz";  
    } else if (fizzy(n)) {  
        return "Fizz";  
    } else if (buzzy(n)) {  
        return "Buzz";  
    } else {  
        return String.valueOf(n);  
    }  
}
```

Green – We are back to a happy place

```
public String value(int n) {  
    if (fizzy(n) && buzzy(n)) {  
        return "FizzBuzz";  
    } else if (fizzy(n)) {  
        return "Fizz";  
    } else if (buzzy(n)) {  
        return "Buzz";  
    } else {  
        return String.valueOf(n);  
    }  
}
```

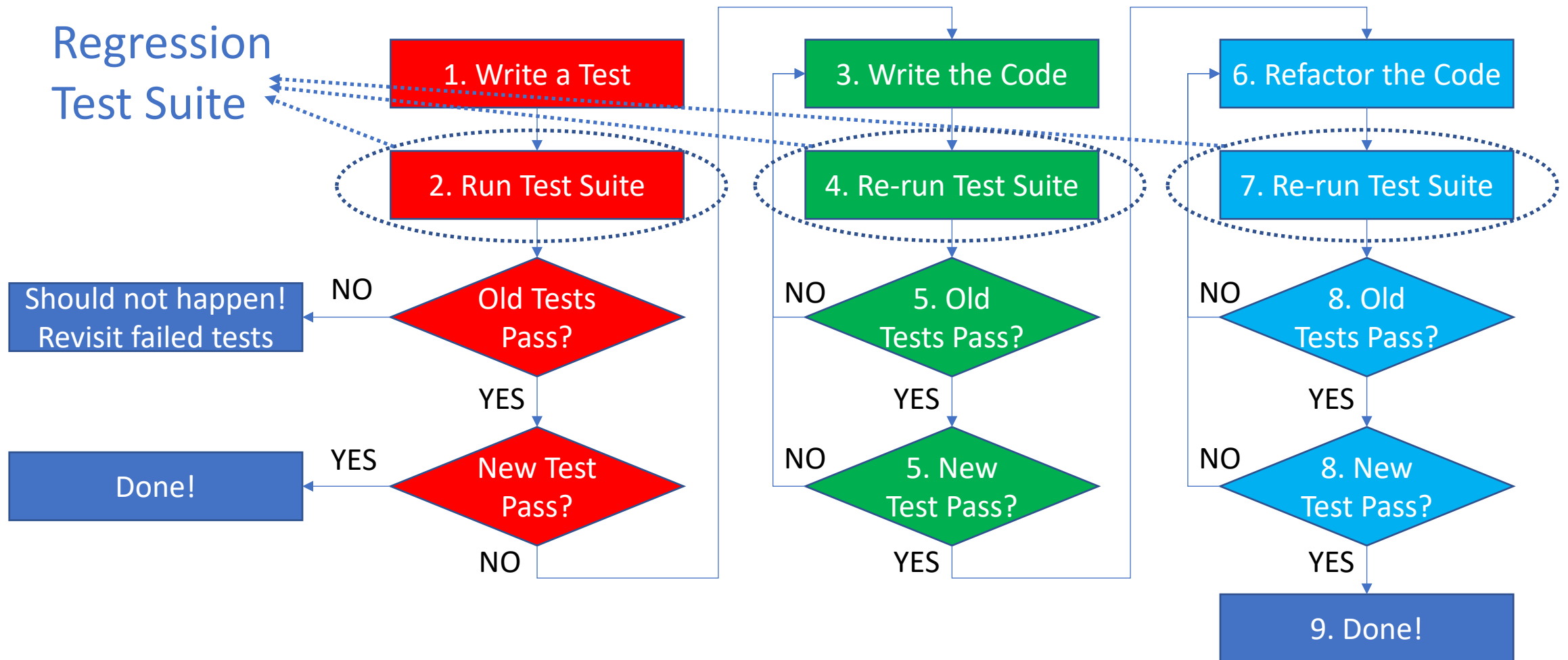


Refactor – Nothing to do

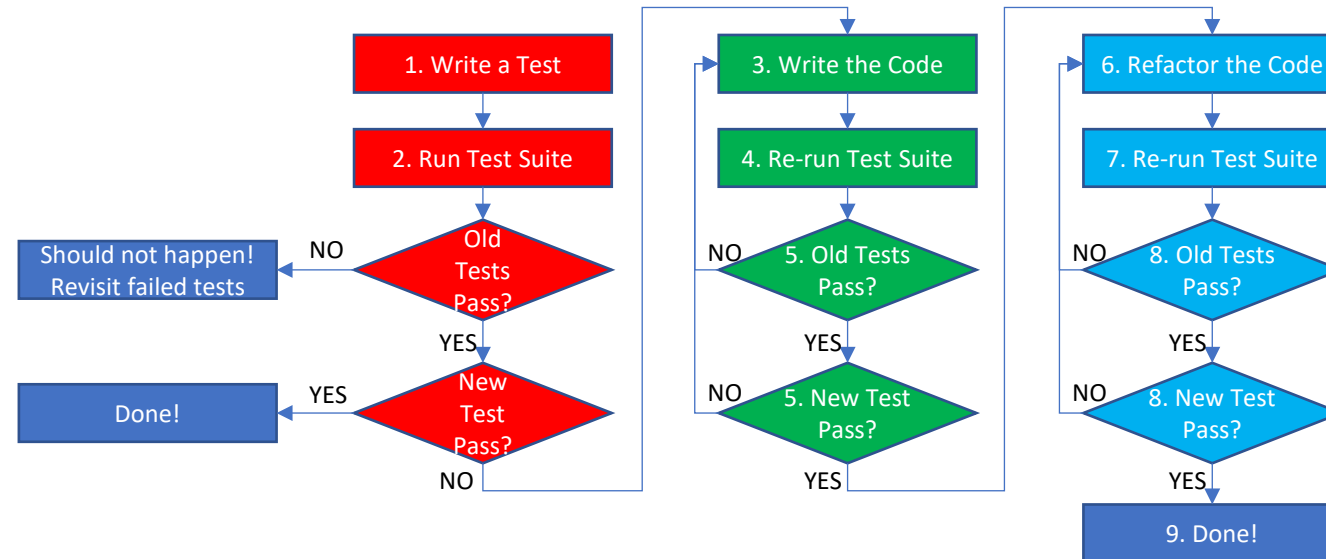
```
public String value(int n) {  
    if (fizzy(n) && buzzy(n)) {  
        return "FizzBuzz";  
    } else if (fizzy(n)) {  
        return "Fizz";  
    } else if (buzzy(n)) {  
        return "Buzz";  
    } else {  
        return String.valueOf(n);  
    }  
}
```



Flow Chart of Red-Green-Refactor Loop



Flow Chart of Red-Green-Refactor Loop



- Write one test at a time to test one behavior
 - Keep a tight turnaround cycle!
- Write just enough code to pass test
 - Resist temptation to go into untested territory
- Refactor at every iteration
 - Focus on correctness in **Green** phase; focus on improvement in **Refactor** phase

Benefits of TDD

1. Development is driven by requirements
 - Requirements drives → Testing drives → Development
 - End-result: all coding effort is tightly bound to the requirements
2. 100% test coverage all the time
 - Test to cover code is written before implementing the code
 - End-result: defects are caught immediately as they are introduced
3. Ensures development happens in small increments
 - Only code sufficient to pass the next test is written at a time
 - End-result: easy to debug since you only need to look at that small increment

Drawbacks of TDD

1. Tests become part of the overhead of the project
 - In terms of maintenance: test code is also code you need to maintain
 - In terms of testing time: sheer testing time slows down **RGR** loop
2. Refactoring and architectural redesigns is often skipped
 - TDD keeps you short-sighted and super-focused on passing tests
 - Good software architecture design is put on the backburner
 - By the time you decide to refactor, the refactor becomes costly
(And guess what, updating test code in step with the refactor itself is costly)

Now Please Read Textbook Chapter 15

- For an alternative view on TDD read the following:
 - “TDD is dead. Long live testing.” - David Heinemeier Hansson:
<https://dhh.dk/2014/tdd-is-dead-long-live-testing.html>
 - “TDD, Where Did It All Go Wrong” - Ian Cooper:
<https://www.youtube.com/watch?v=EZ05e7EMOLM>
 - Like everything else in software QA, TDD is a tool not a religion